

Interpreting “Liberation Day” stock returns through the lense of international trade theory *

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Abstract

This paper analyses stock market reactions to the “Liberation Day” import tariffs announced on 2 April 2025. Motivated by theories of international trade, we regress abnormal returns on the industry-level exposure to import tariffs, as well as a measure of input cost shocks, capturing how the tariffs affected firms’ costs of imported intermediate inputs. Consistent with theories of international trade, we find that higher tariffs on firms’ imported intermediate inputs are associated with negative abnormal returns. Surprisingly, the direct impact of protective import tariffs is also negative. We show that this is driven by industries in which firms have a larger export share of total output, suggesting that the negative abnormal return reaction captures expectations of future retaliatory tariffs.

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1 Introduction

The so-called “liberation day” import tariffs announced by the Trump administration in April 2025 raised US import tariffs to their highest level since the 1930s. US stock markets reacted with steep declines in share prices, suggesting that investors anticipated the tariffs to lead to lower future profits of listed companies.

In this paper, we analyse this stock market reaction in the light of theories of international trade. We believe that this is interesting for at least two reasons. First, while a number of papers have already analysed stock market reactions to the liberation day tariffs, those analyses have been only loosely informed by theories of international trade within textual analyses of 10K filings. As we’ll show, using theories of international trade to inform the analysis of abnormal stock market returns - especially a theory-consistent input channel - helps provide a more complete picture of stock market return patterns.

Second, theories of international trade themselves can be evaluated in light of stock returns. Given that stock prices aggregate all information available to investors at a given point in time, stock market reactions to a well-defined trade-policy event can tell us whether the predictions of trade theories are in line with the expectations of rational investors, hence providing insights into the empirical validity of these theories.

We start our analysis by estimating abnormal returns of S&P 500 stocks on 3 April 2025, the day after the Liberation day tariffs were announced. We then interpret the estimated abnormal returns in light of theories of international trade. A basic prediction of international trade theory is that import tariffs will raise domestic prices and hence increase the profits of the domestic firms producing these goods. This suggests that the liberation day tariffs should be associated with positive abnormal returns. However, higher tariffs will also raise the costs of imported intermediate inputs, raising the production costs of US firms and hence lowering their profits.

We investigate the extent to which these two channels were active on 3 April 2025 by regressing abnormal returns on import tariffs as well as on intermediate input tariffs, with the latter being calculated by weighing import tariffs by sectoral input usage taken from US BEA input-output tables. Our regression results confirm the hypothesis that higher intermediate input tariffs lead to lower profits. Firms in sectors with bigger increases in intermediate input tariffs saw significantly lower abnormal returns. We also show that this effect is larger when the industry’s imports are larger share of its inputs. In terms of magnitudes, a 10% increase in tariffs on imported inputs reduces abnormal returns by 0.72% for the average firm.

For final goods, surprisingly, we estimate that higher import tariffs also had a negative effect on abnormal returns, even when controlling for intermediate input tariffs, seemingly contradict-

ing basic predictions of international trade models. For the average firm, a 10% tariff leads to a 1.1% reduction in abnormal returns. We postulate that this surprising negative reaction was due to US investors anticipating retaliatory tariffs of the US' major trading partners, which would have harmed US exports and hence profitability. To investigate this hypothesis, we construct a simple proxy for the likely impact of retaliatory tariffs, which is the interaction of import tariffs with the share of exports in the total output of firms' primary industries. When we include this interaction in our baseline regression, we find the interaction to be negative and statistically significant, whereas the import tariff on its own is no longer significant. This suggests that the negative impact of import tariffs in our baseline regression was driven by export-oriented industries, lending support to our hypothesis that the threat of retaliation weighed negatively on US stock prices.

Finally, we compute overall abnormal return reactions using our full set of regressors, finding that the liberation day tariffs generated negative abnormal returns of -5.4%, of which 22% was due to higher costs of intermediate inputs, and 14.8% due to expectations of retaliatory tariffs. The great majority of the remaining variation is unexplained, and likely due to increased market uncertainty.

Our paper contributes to three strands of the literature. A number of papers have already estimated abnormal return reactions to the announcement of the liberation day tariffs through textual analyses of 10K filings (Hu and Han (2026); Coccozza and Gallo (2025); or Al-Kharusi and Elroukh (2025)). In contrast to these papers, we use international trade theory to precisely define covariates that may impact abnormal returns, revealing additional interesting patterns not analyzed in existing studies. We include as a control variable a text-based measure of tariff risk in 10K filings, as calculated in the existing literature, and we find similar magnitudes for this variable, and that our results are robust to its inclusion.

Second, we contribute to a small literature looking at stock price reactions to trade policy events. Hartigan et al. (1986, 1989), Hughes et al. (1997), Blonigen et al. (2004) and Crowley et al. (2019) look at stock price reactions to sector-specific anti-dumping duties. Grossman and Levinsohn (1987) use stock price reactions to test the specific factors model of international trade, and Breinlich (2014) uses stock market reactions to the Canada-US free trade agreement of 1989 to test predictions of heterogeneous firm models. In contrast to this prior literature, we evaluate stock market reactions to the Liberation Day tariffs through the lens of two central mechanisms in modern trade theory: the protective effect of direct output tariffs versus the cost-inflating effect of intermediate input tariffs. Our findings show that stock market reactions to the liberation day tariffs were in line with the input cost channel, but not with the protection

channel. Rather, our results suggest that direct protection was associated by investors with raising the likelihood of retaliatory tariffs, further decreasing future expected profits.

The rest of this paper is structured as follows. Section 2 describes our stock price data and how we estimate abnormal returns. Section 3 outlines basic predictions of modern quantitative trade models and how we aim to measure them empirically. Section 4 correlates these measures with our estimated abnormal returns and interprets the results in light of trade theory. Section 5 concludes.

2 Methodology and Data

We estimate abnormal returns using all firms listed in the S&P 500 as our sample. We first estimate a model of “normal” stock returns which adjusts for differences in risk and other characteristics of stocks. A standard approach in the literature is to use a “market model” which relates the return, r_{it} , on stock i at time t to a stock-specific constant, α_i , and the return on the market portfolio, R_{mt} :

$$r_{it} = \alpha_i + \beta_i R_{mt} + e_{it}, \quad t \in T_1 \quad (1)$$

where $r_{it} = \ln(\text{price}_{i,t}) - \ln(\text{price}_{i,t-1})$, e_{it} is the mean-zero random component of the return-generating process and T_i is the so-called pre-event window of stock price data on which equation 1 is estimated. This method controls for differences in average returns across stocks (α_i), a stock’s (non-diversifiable) risk as measured by β_i and movements in the market portfolio.

On event dates, stock returns also have an “abnormal” component, τ_{it} , which in the present context could be caused by the arrival of unexpected news about future import tariffs. Thus, on event dates, stock returns are given by:

$$r_{it} = \alpha_i + \beta_i R_{mt} + \tau_{it} + e_{it}, t \in T_2 \quad (2)$$

where T_2 denotes the event window (in our case, 3 April 2025). Having obtained estimates of α_i and β_i using stock price data from the pre-event window only, we compute abnormal returns estimates, $\hat{\tau}_{it}$, as a prediction error for the event window:

$$\hat{\tau}_{it} = r_{it} - \hat{r}_{it} = r_{it} - \hat{\alpha} - \hat{\beta}_i R_{mt}, t \in T_2 \quad (3)$$

where the predicted values, $\hat{r}_{it}$, have been constructed using the pre-event window estimates of α_i and β_i and the return on the market portfolio on the event day.

We estimate abnormal returns for all stocks in the S&P 500, using the year 2024 as our pre-event window, and 2-3 April 2025 as our event window. Data for stock prices are obtained

from Compustat North America, accessed via the Wharton Research Data Services. We use the MSCI World ETF as our proxy for the market portfolio and obtain returns on this ETF from investing.com.

Once we have estimated abnormal returns, we then regress them on proxies for different mechanisms highlighted in international trade theory. An important issue for inference in event studies is the correct computation of standard errors. Following [Karafiath \(1994\)](#) and [Harrington and Shrider \(2007\)](#), we estimate our model via ordinary least squares (OLS) with heteroskedasticity robust standard errors. To account for possible cross-sectional dependence, we also cluster standard errors by industry throughout this paper.

3 Trade theory predictions and measurement

Our working hypothesis is that the liberation day tariffs should affect market expectations separately through intermediate inputs and final output. For inputs, tariffs will make production more costly for firms that use imported inputs, reducing expected returns. These effects could be further magnified if imported intermediate inputs are used as inputs in other intermediate goods. To account for such second-round effects, we will utilize a measure of input tariffs that weighs import tariffs by sectoral input usage taken from US BEA input-output tables (see below for details).

Second, we also focus on the protective effects of tariffs. A tariff on an imported good will raise the domestic price of that good. This should benefit domestic firms in that industry, raising expected profits ([Ossa \(2014\)](#)). However, many of these firms that are protected by tariffs may also be exporters, and subject to retaliatory tariffs by US trading partners. Thus, retaliation may reduce expected returns if investors find significant retaliation likely.

To measure these channels, we start by defining each industry’s tariff, not accounting for input use. Because the Liberation Day tariffs were country-specific, we construct a weighted average of the country-specific tariffs, where the weights reflect the country composition of imports in each industry. Specifically,

$$\text{IndustryTariff}_{i,US} = \sum_k \hat{t}_k \times \text{CountryImportShare}_{i,k} \equiv \text{INDT}_i \quad (4)$$

where $\hat{t}_k$ denotes the tariff imposed by the United States on imports from country k , and $\text{CountryImportShare}_{i,k}$ is the share of industry i ’s imports that originate from country k . This measure captures the average tariff imposed on imports competing with domestic producers in industry i .

To measure tariff exposure through inputs, we use the 2017 BEA Input-Output Use Table

to measure industries' input sourcing patterns. Following the methodology of [Breinlich et al. \(2026\)](#), we construct an input-linkage-adjusted tariff measure as

$$\text{InputTariff}_{i,US} = (I - \Gamma_U)^{-1} \Gamma_U \cdot \text{IndustryTariff}_{i,US} \equiv \text{INPT}_i \quad (5)$$

where Γ_U is the matrix of input-output coefficients constructed from the 2017 BEA Use Table. The Leontief inverse $(I - \Gamma_U)^{-1}$ propagates tariff shocks through upstream production networks, allowing the measure to capture both direct and indirect tariff exposure arising from intermediate input linkages.

Because Γ_U does not distinguish between domestic and imported inputs, we also construct an industry-level variable, *ImportShare_i*, which is the ratio of imported inputs to total inputs for industry *i* in 2017. To examine whether tariff exposure is amplified by reliance on imported inputs, we interact the input tariff measure with the industry's import share.

Then, to examine whether stock market reactions reflect concerns about foreign retaliation, we construct an industry-level export-intensity measure, *ExptoOutput_i*, which is the ratio of domestic exports to total industry output in 2017. Industries that export a larger fraction of their output are more exposed to potential retaliatory tariffs and reductions in foreign demand.¹

Finally, we also construct the text-based tariff exposure measure used in [Hu et al. \(2026\)](#) and [Andreou et al. \(2026\)](#), where we collect 10-K filings for *S&P* 500 firms and identify paragraphs containing tariff-related keywords. Within these paragraphs, we count mentions of each country listed in the Liberation Day tariff annex. This approach yields a revealed measure of firms' tariff exposure based on their own disclosures.

4 Explaining abnormal returns

Table 1 presents summary statistics for all variables used in the analysis. The final sample consists of 420 firms out of the 503 *S&P* 500 firms. The loss of observations is primarily due to the failure to match some firms' NAICS industry classifications to the corresponding BEA benchmark input-output industry codes required to construct the industry-level tariff exposure measure.²

Overall, *S&P* 500 firms experienced negative abnormal returns on 3 April 2025; approximately -5% abnormal return at the mean/median, with significant variation around this point. In terms of tariffs, we find that the average industry-level import incurs a 7.7% tariff, but that this varies from 0% and 32.7%. When these tariffs are cycled through the input-output tables,

¹Industry-level export data are obtained from USA Trade Online

²Because the 10-K filing for Hess Corporation is unavailable, the text-based tariff exposure measure is available for only 419 firms.

the average industry faces a 12% tariff on inputs, and a max of 42%.

We next explain the cross-sectional pattern of these returns using measures of input and output exposure:

$$\begin{aligned}\hat{\tau}_{ij} = & \kappa + \gamma_1 INPT_j + \gamma_2 INDT_j + \gamma_3 ExptoOutput_j + \gamma_4 INDT_j \cdot ExptoOutput_j \\ & + \gamma_5 ImportShare_j + \gamma_6 INPT_j \cdot ImportShare_j + \gamma_7 TextTariffExposure_{ijt} + \mu_{ij},\end{aligned}\quad (6)$$

where $\hat{\tau}$ are the estimated abnormal returns for firm i belonging to industry j , $INPT_j$ denotes the intermediate input tariff of industry j , $INDT_j$ the import tariff of industry j , $ExptoOutput_j$ is the export to output ratio for industry j , $ImportShare_j$ is the import share of intermediate inputs in industry j , and $TextTariffExposure_{ijt}$ is the exposure to tariffs calculated using 10K filings as used in the existing literature.

The results of estimating this regression are presented in Table 2. In the first column, we include only the input tariff and the industry tariff. Here, the results show that intermediate input tariffs have a negative association with abnormal returns on 3 April 2025, with a one percentage point higher input tariff lowering abnormal returns by about 0.1%. As previously discussed, this is consistent with standard quantitative trade models which predict that higher tariffs on intermediate inputs increase importing firms' costs and hence lower their profits. At the sample average tariff, the typical firm experiences a 1% lower abnormal return through this channel.

More surprisingly, industry tariffs also show a negative association with abnormal returns. Specifically, the average firm experiences a 0.77% reduction in abnormal returns through this channel. This result seems to contradict the basic prediction of trade models that more import protection should benefit domestic firms by allowing them to charge higher prices. Instead, the result suggests that investors were already anticipating retaliatory tariffs by foreign countries that may impact larger, exporting firms. To test this, we interact our import tariff regressor with the export intensity of firms' industries, with the results presented in column (2) of Table 2. The coefficient on this new interaction is negative and statistically significant, while the coefficient on the original import tariff regressor is no longer significant. This suggests that the negative impact we had estimated in column (1) was indeed driven by firms in export-intensive industries, who stood to lose from potential future retaliatory tariffs.

In column (3), we add $ImportShare_j$, and an interaction with $INPT_j$, to evaluate whether more tariff exposure to imported intermediates leads to a lower abnormal return. Indeed, this is what we find; while the coefficient on import share is insignificant, the interaction of import share and the input tariff is negative and significant. Further, note that the coefficient on input

tariff is now very small and insignificant. Thus, we conclude that being more exposed to the input tariff through imported intermediates leads to more negative returns.

Lastly, we extend our results thus far with a common variable from the existing literature, which is the text based measure of tariff exposure, calculated by using firm’s 10K filings. The results of this addition are in column (4), where there is a negative impact of this exposure measure on abnormal returns, and similar in magnitude with existing studies in [Hu et al. \(2026\)](#) and [Andreou et al. \(2026\)](#).

4.1 Decomposition

We decompose the average fitted abnormal return into the average contribution of each regressor, where each contribution equals the estimated coefficient multiplied by the sample mean of the corresponding regressor. We then report each contribution as a share of the average fitted abnormal return, as well as its absolute share among all fitted components. Specifically:

$$C_k = \hat{\beta}_k \bar{X}_k \quad (7)$$

$$\text{Share}_k = \frac{C_k}{\overline{AR}} * 100 \quad (8)$$

$$\text{AbsShare}_k = \frac{|C_k|}{\sum_k |C_k|} * 100 \quad (9)$$

These results are presented in Table 3. Here, we see that the regression constant captures the greatest share of abnormal returns, suggesting that there are large, aggregate forces at play that impact the abnormal returns of S&P 500 companies beyond a normal market index. Of the rest of the variables, we see that the interaction terms, and the import share variable, explain a large degree of the abnormal returns. Text based exposure measures account for some as well, but far less than other industry-level variables that are more centered on trade theory.

5 Conclusions

This paper has analysed to what extent stock market reactions to the liberation day tariffs were consistent with quantitative trade models. We find strong evidence that US investors did indeed expect negative consequences from higher intermediate input tariffs, which trade models predict should increase firms’ costs and lower profits. By contrast, we also find a negative impact of import protection on abnormal returns, which at first sight is at odds with the predictions of quantitative trade models. However, we showed that this is likely due to those models abstracting from retaliatory tariffs, which we have argued can explain the negative effect of import tariffs on abnormal return. We thus conclude that standard quantitative models of international trade

can be helpful in explaining stock market reactions to major trade policy events, but in order to provide a complete picture of mechanisms during a trade war, they need to be augmented to allow for the possibility of retaliatory tariffs of foreign trade partners, and the subsequent effects on firm profits (as in [Ossa \(2014\)](#)).

Table 1: Summary Statistics

Variable	Mean	Std. Dev.	Median	Min	Max	N
Abnormal Return	-0.0545	0.0559	-0.0523	-0.2180	0.0923	420
Input Tariff	0.1210	0.0794	0.0993	0.0000	0.4175	420
Industry Tariff	0.0773	0.0966	0.0000	0.0000	0.3274	420
Export-Output Ratio	0.1361	0.2348	0.0000	0.0000	1.5629	420
Import Share of Intermediates	0.1398	0.1849	0.0309	0.0000	0.7271	420
Text Tariff Exposure	0.1016	0.1519	0.0000	0.0000	0.3822	419

Table 2: Abnormal Returns and Tariff Exposure

	(1)	(2)	(3)	(4)
INPT	-0.1139*** (0.0404)	-0.0982*** (0.0365)	-0.0130 (0.0629)	-0.0146 (0.0627)
INDT	-0.1099** (0.0453)	-0.0094 (0.0545)	-0.0187 (0.0653)	0.0057 (0.0630)
ExptoOutput		-0.0061 (0.0341)	-0.0009 (0.0330)	-0.0021 (0.0330)
INDT $\times$ ExptoOutput		-0.2217* (0.1277)	-0.2360* (0.1231)	-0.2415* (0.1247)
ImportShare			0.0659 (0.0633)	0.0668 (0.0630)
INPT $\times$ ImportShare			-0.4359* (0.2480)	-0.4210* (0.2491)
Text tariff exposure				-0.0343* (0.0176)
Constant	-0.0322*** (0.0081)	-0.0352*** (0.0080)	-0.0443*** (0.0095)	-0.0426*** (0.0093)
Observations	420	420	420	419
R^2	0.079	0.107	0.119	0.126
Adj. R^2	0.075	0.098	0.106	0.111

Notes: Standard errors clustered at the NAICS industry level (170 industries) are reported in parentheses. INPT denotes input tariff exposure, and INDT denotes industry tariff exposure. The reported significance levels are based on the t -distribution with cluster-robust standard errors. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Decomposition of Predicted Abnormal Returns

Variable	Contribution (C_k)	$Share_k$ (%)	$Abs. Share_k$ (%)
INPT	-0.0018	3.247	2.391
INDT	0.0004	-0.807	0.594
ExptoOutput	-0.0003	0.521	0.383
ImportShare	0.0093	-17.075	12.577
INDT $\times$ ExptoOutput	-0.0064	11.771	8.670
ImportShare $\times$ INPT	-0.0096	17.707	13.042
Text Exposure	-0.0035	6.400	4.714
Constant	-0.0426	78.237	57.627

Notes: This table presents the decomposition described in section 4.1, equations 7-8.

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